

6	(a) force due to magnetic field is constant force is (always) normal to direction of motion this force provides the centripetal force	B1	A1	[3]
	(b) $mv^2 / r = Bqv$ hence $q / m = v / Br$	M1	A0	[1]
	(c) (i) $q / m = (2.0 \times 10^7) / (2.5 \times 10^{-3} \times 4.5 \times 10^{-2})$ $= 1.8 \times 10^{11} \text{ C kg}^{-1}$	C1	A1	[2]
	(ii) sketch: curved path, constant radius, in direction towards bottom of page tangent to curved path on entering and on leaving the field	M1	A1	[2]